

Overview

1. What is The Carpentries
2. A Very Brief History of The Carpentries
3. Where does HPC Carpentry fit?
4. Current Status
5. Roadmap and Challenges
6. The mini version
7. Comms Channels

What we will look at is what The Carpentries is about and how it came into existence. Most of you are probably quite familiar with it anyway so it will be very short. We'll then look at what HPC Carpentry is and how that came about. What the current status of the HPC Carpentries is and what our future plans are. I wouldn't be myself if I didn't squeeze a bit of the miniHPC in here, so I'll quickly mention that and then I'll leave you with ways to get into contact us so that you too can join us in bringing HPC Carpentries to researchers.

└ A Very Brief History of The Carpentries

It all started back in 1998 with Greg Wilson.

- Software Carpentry was founded in 1998 with the mission of teaching job skills for research computing by Greg Wilson and Brent Gorda.



Dr Greg Wilson
Software Carpentry Co-founder

└─Where does HPC Carpentry fit?

1. Knowing about version control and functions does not quite solve all problems.
2. Researchers with HPC requirements do not necessarily know how to effectively scale their applications
3. HPC facility operators become frustrated by non-knowledgeable users making use of shared resources.
4. Solution: Use Carpentries techniques.

1. So once you write using functions and your code is in GitHub, you try running it on an HPC and you might find it runs a bit faster because it just so happens that the login node has a better spec than your average laptop or desktop. But people start yelling at you for running stuff on the login node and you don't know what the problem is because you've never heard about the Slurm scheduler thing and you neither does MPI sound familiar
2. Using the Carpentries' style and pedagogy we HPC Carpentry address this problem

└─ How did it all begin

- ▶ Earliest commit to hpc-carpentry GitHub 2013 by Compute Canada
- ▶ Peter Steinbach blog post (2017) - HPC in a day
- ▶ CarpentryCon 2018, 2020, 2022
- ▶ Super Computing BoF 17, 18, 19, 21
- ▶ Winter 2021, accepted into Carpentries Incubator
- ▶ Begin working towards Lesson Program Incubation in 2021 and formally accepted to the Lesson Program Incubation in June 2023
- ▶ April 2024, workflow lessons enters testing

1. The earliest commit I could find in the hpc-carpentry organisation was 2013. I watched a recording of a presentation given by Andrew Reid at SIGHPC about a year ago to get an idea of the history of HPC Carpentry and these were some of the events that Andrew mentioned that were significant in the spread of the word of HPC Carpentry, by talking to people, running workshops and getting feedback.

└ Where are we now?

1. We are doing things in the Carpentries Way
 - ▶ two day setting
 - ▶ students type along with instructors
 - ▶ get feedback
2. Two main takeaways:
 - ▶ muscle memory
 - ▶ vocabulary
3. Not experts after two days but:
 - ▶ know what answer looks like
 - ▶ know how to find answers

1.

└─ Current Status: Lesson Program Adoption

- ▶ Adoption as an official lesson program of The Carpentries was planned for 1 September 2026
- ▶ Changes in Carpentries Core Team support called for a short delay
- ▶ Adoption will be completed between now and November 2026

Those of you who attended my talk last year might recall that the big news back then was that HPC Carpentry would be adopted by The Carpentries, becoming their fourth official lesson program. The launch date for the new program was set for 1st September this year, which was last week. Unfortunately, that timeline had to be adjusted earlier this summer, after The Carpentries announced a reduction in Core Team capacity. We were all really keen to make it happen, though, and most of the work for the adoption had already been done, so the delay should be quite short. The deadline is now 1 November this year, and we are optimistic that we will be able to complete the process before then. Watch out for further announcements on The Carpentries blog.

└ Your Help is Needed

Ask for suggestions for domain-specific conferences where we could reach members of HPC Carpentry's target audience of learners and/or potential instructors/hosts for workshops.